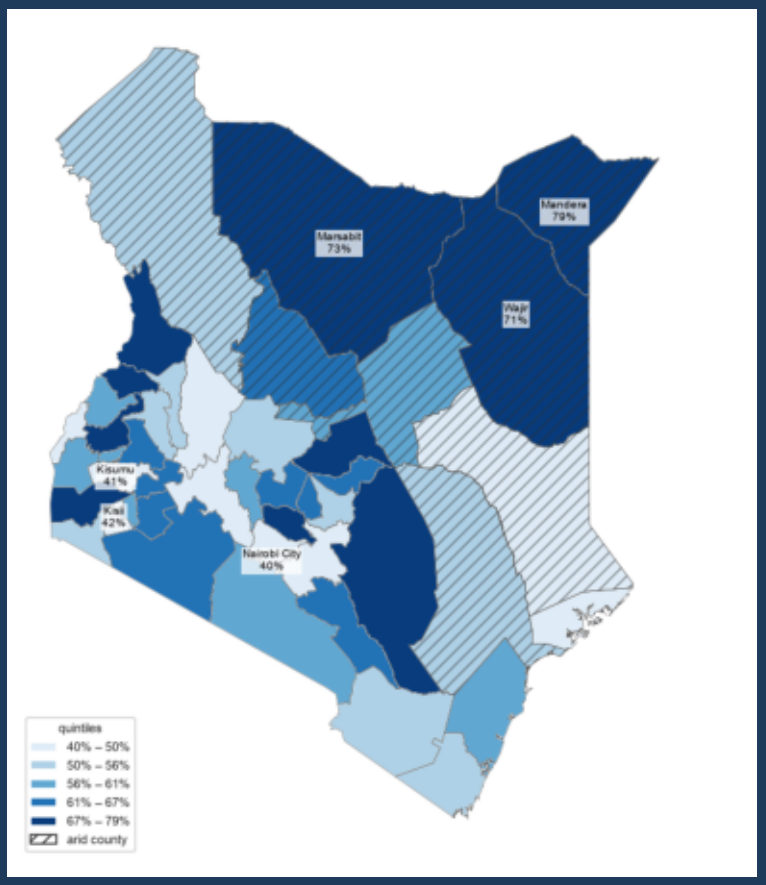
ECONOMETRICS IN ACTION

12 YEARS OF DEVOLUTION · 47 COUNTIES · KES 4.36 TRILLION

Does budget absorption predict development?

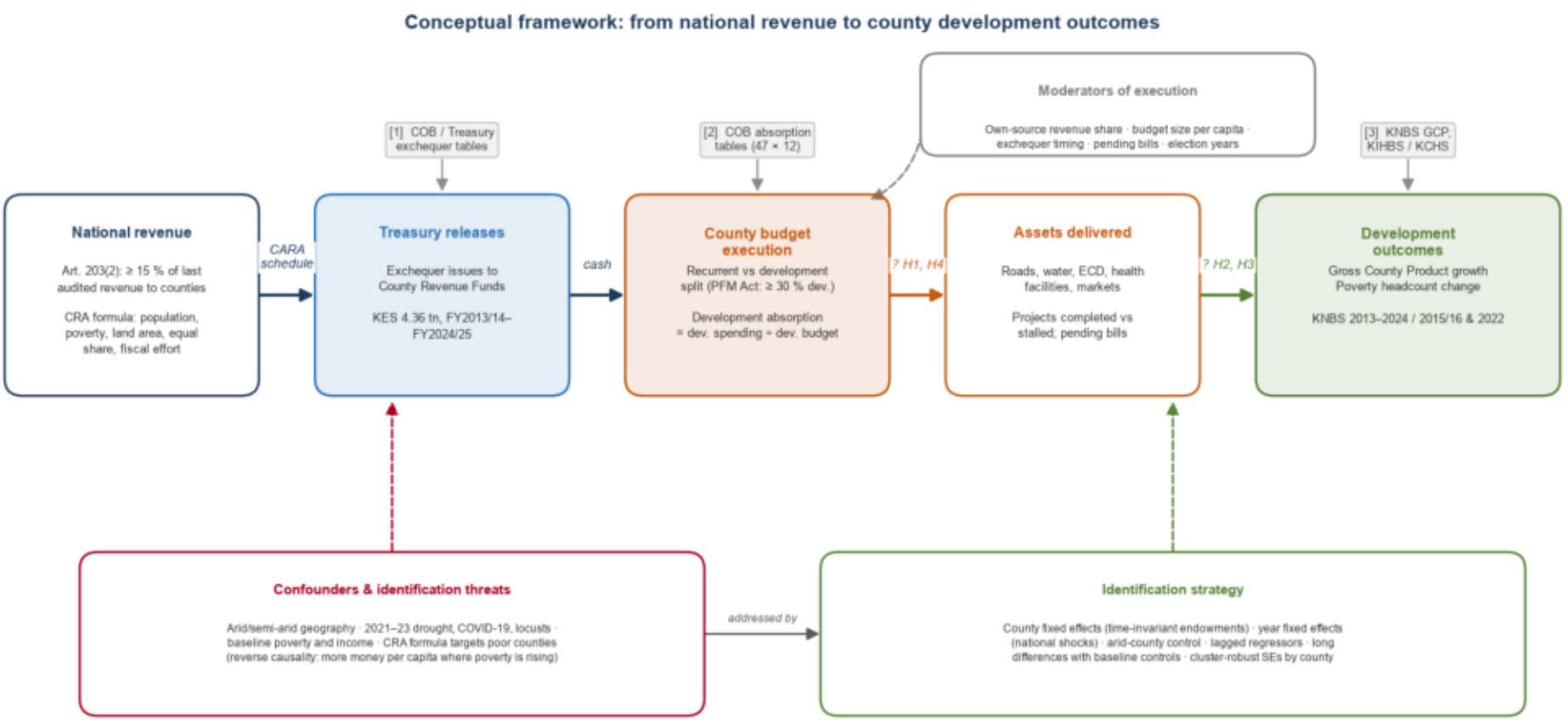
What Kenya's county absorption rates do — and do not — tell us about growth, poverty and project delivery



Peter Elman · 3 November 2026 · swipe →

WHY THIS MATTERS

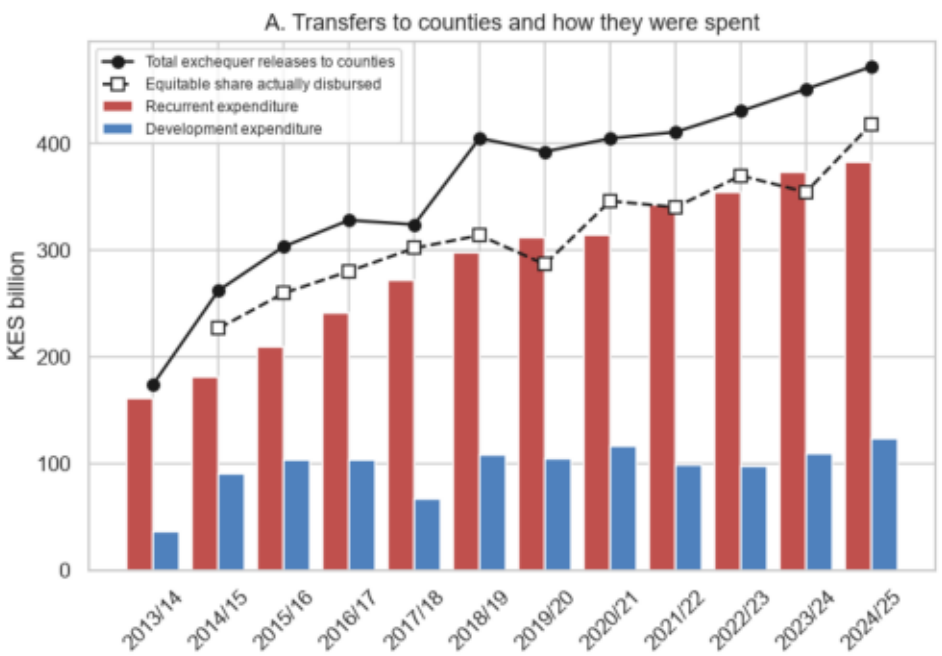
The question

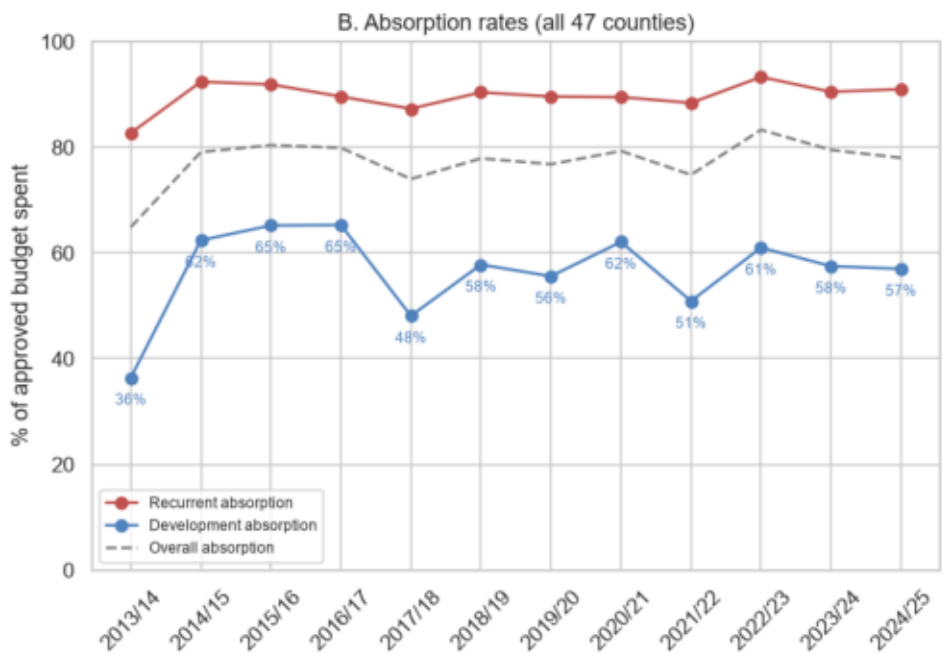


The chain can weaken at [2] (budgeted funds not executed) or between [2] and [3] (execution without measurable welfare change). Shaded boxes are directly observed; H1–H4 test the marked links.

- Every year the Controller of Budget reports that counties spend only ~half of their development budgets. The "absorption rate" has become the headline measure of county performance.
- But does executing a higher share of the development budget actually translate into development — faster Gross County Product growth, lower poverty, completed projects?
- I built a 47-county × 12-year panel (FY2013/14–2024/25) from COB, CRA, Treasury and KNBS data and tested it with fixed-effects panel regressions, a cluster bootstrap and a neural-network benchmark.

The money is real — and growing

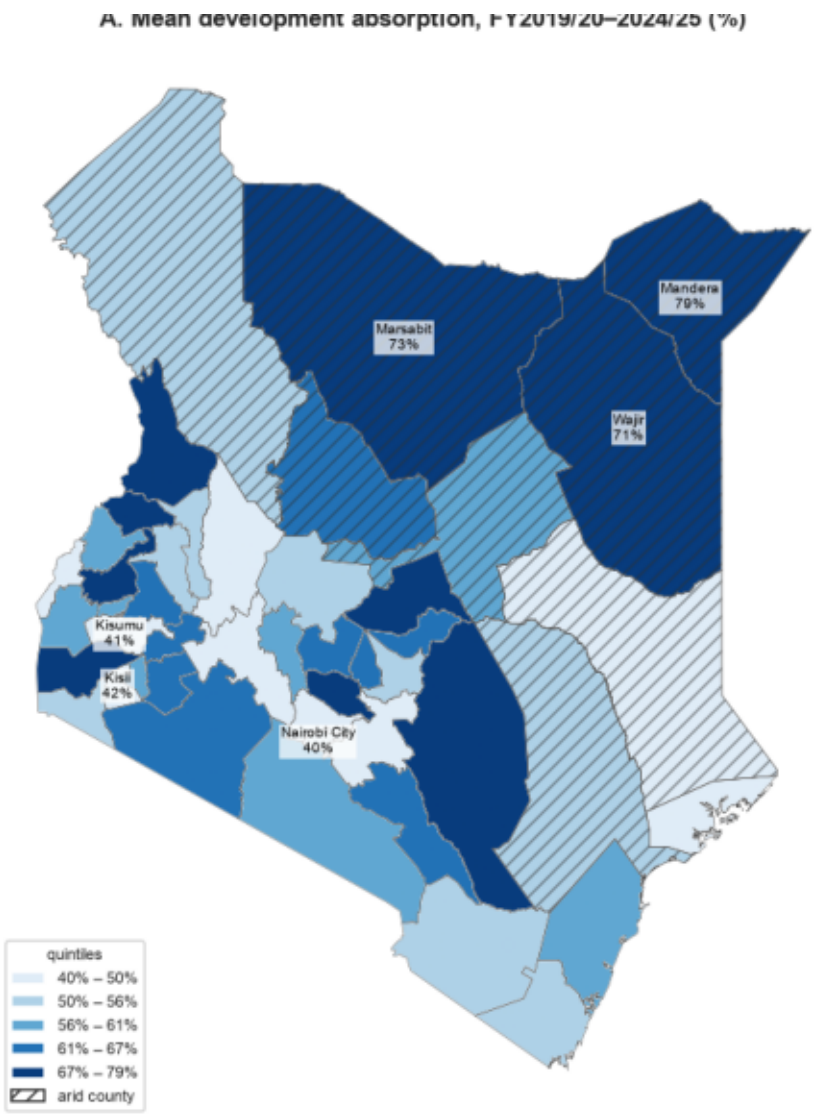


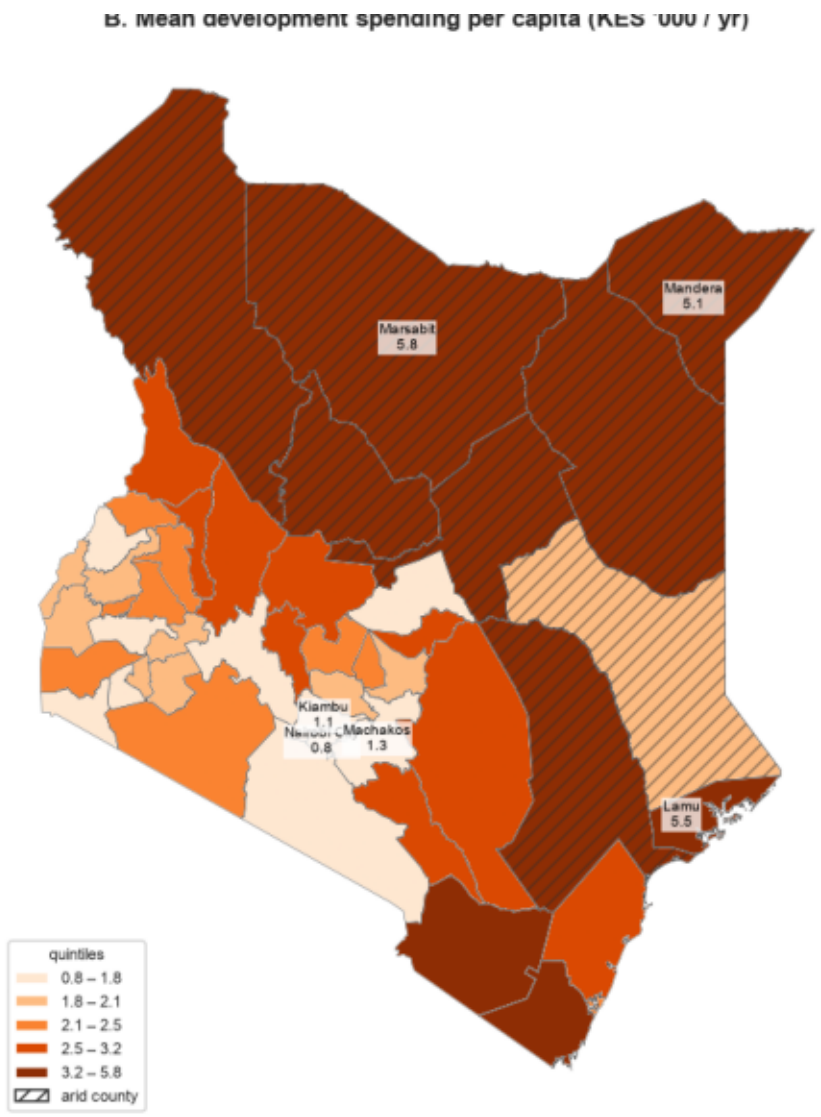


- KES 4.36 trillion released to counties since FY2013/14; releases exceeded county spending in all 12 years.
- Only 27 % went to development (22–26 % in recent years, below the PFM Act's 30 % threshold). Recurrent spending rose from KES 161 bn to 383 bn; development spending barely moved.
- Development absorption: 36 % → 65 % → 48 % (election year) → ~57 %. No trend in a decade. Recurrent absorption: 83–93 % throughout.
- KES 867 bn of approved development budgets went unspent.

Source: Controller of Budget, Annual County Budget Implementation Review Reports FY2013/14–FY2024/25

Absorption follows the core–periphery map



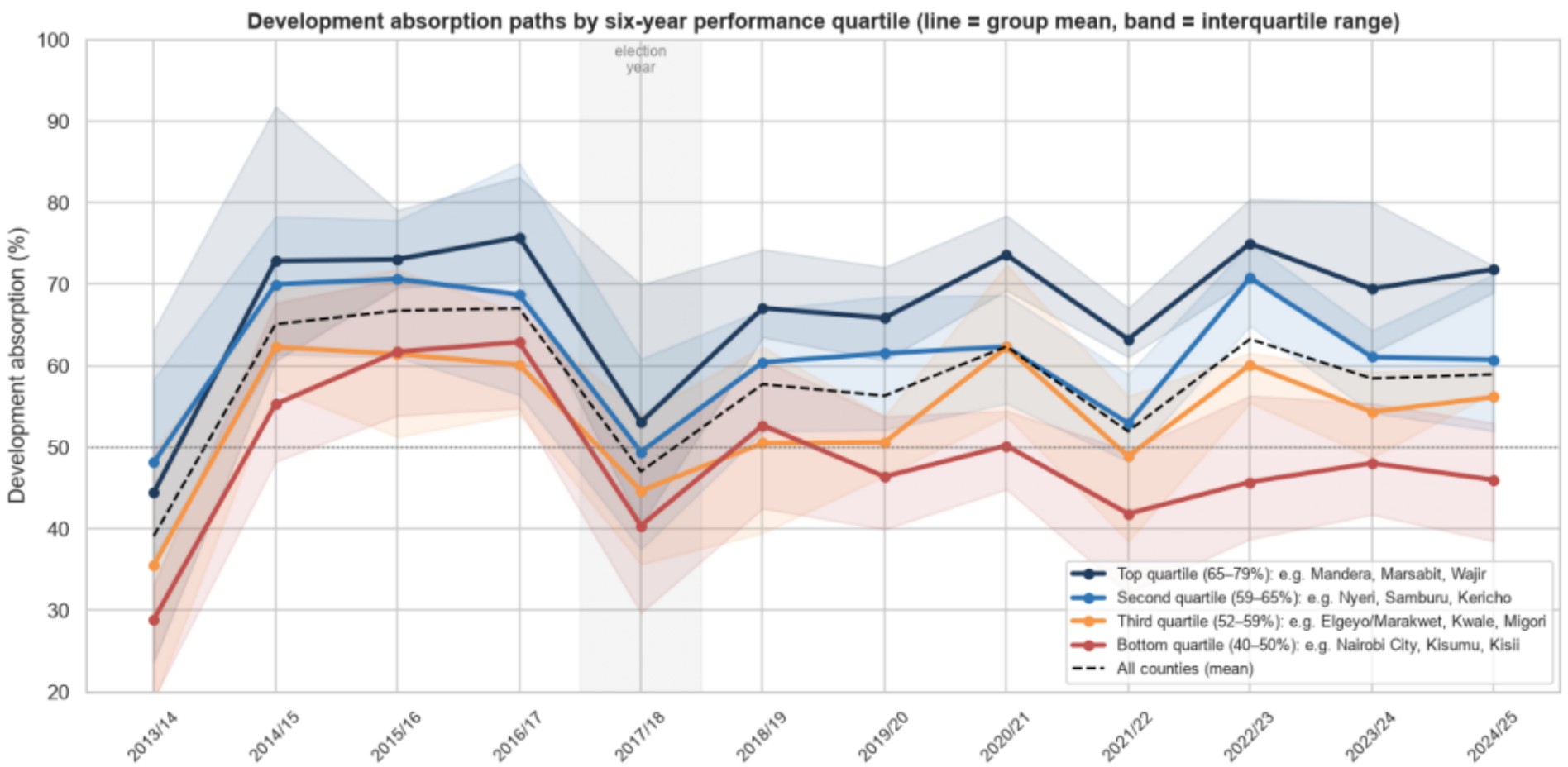


- Highest six-year absorption: Mandera 79 %, Marsabit 73 %, Wajir 71 %. Lowest: Nairobi City 40 %, Kisumu 41 %, Kisii 42 %, Nakuru 44 %.
- The arid north receives the most development funding per person — and executes the highest share of it.
- In absolute terms the unspent budget is concentrated in the big urban counties: Nairobi City KES 36 bn, Nakuru KES 33 bn over six years.

Boundaries: geoBoundaries ADM1 (public domain). Hatched = 8 arid counties.

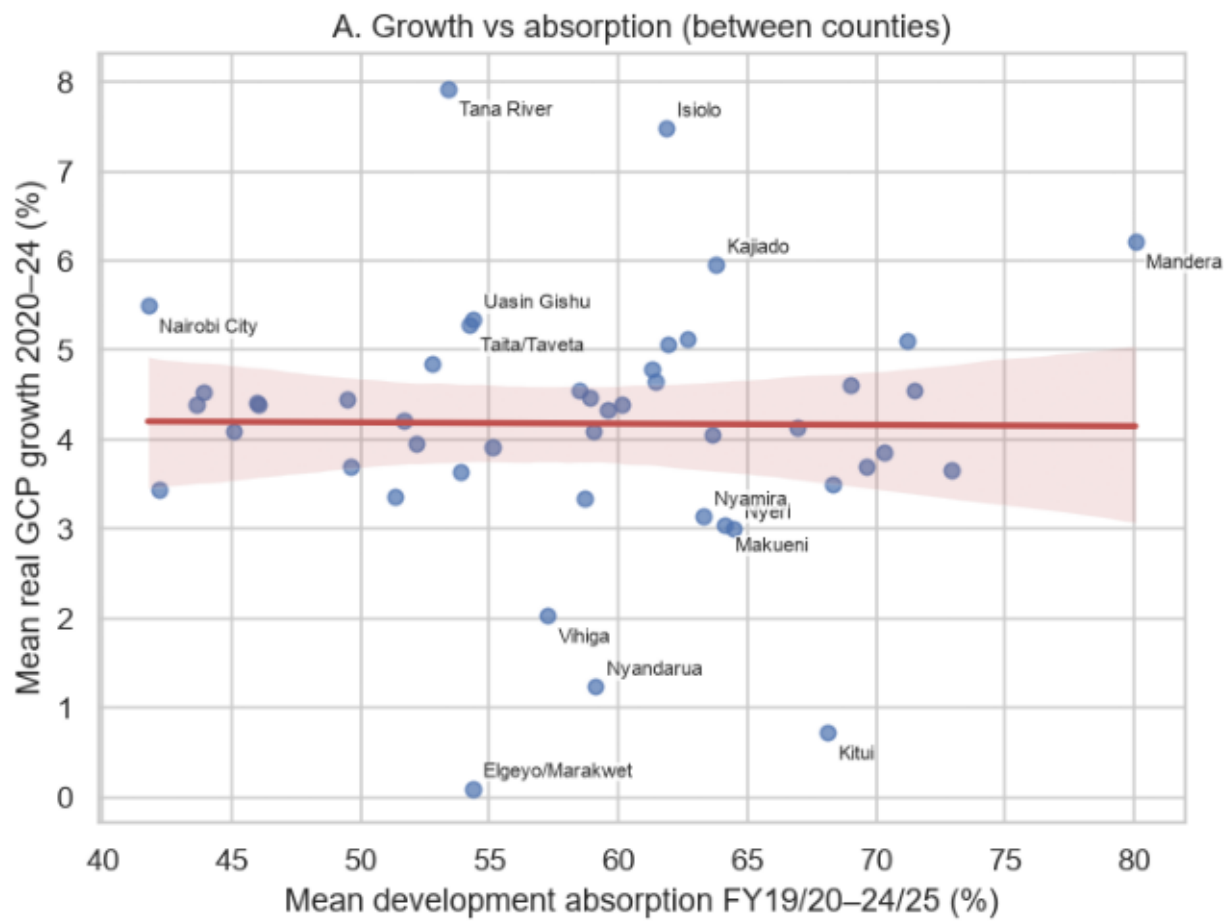
FINDING 3 · PERSISTENCE

Mostly noise, not capacity



- Only 40 % of the variation in absorption is between counties; 60 % is year-to-year movement within the same county. AR(1) = 0.27.
- Every group moves to the same national rhythm — election-year trough, 2021/22 dip — pointing to cash-flow timing and procurement cycles rather than stable institutional quality.
- The groups diverged: the top quartile rose from 64 % to 70 % between the two halves of devolution; the bottom quartile fell from 50 % to 46 %.

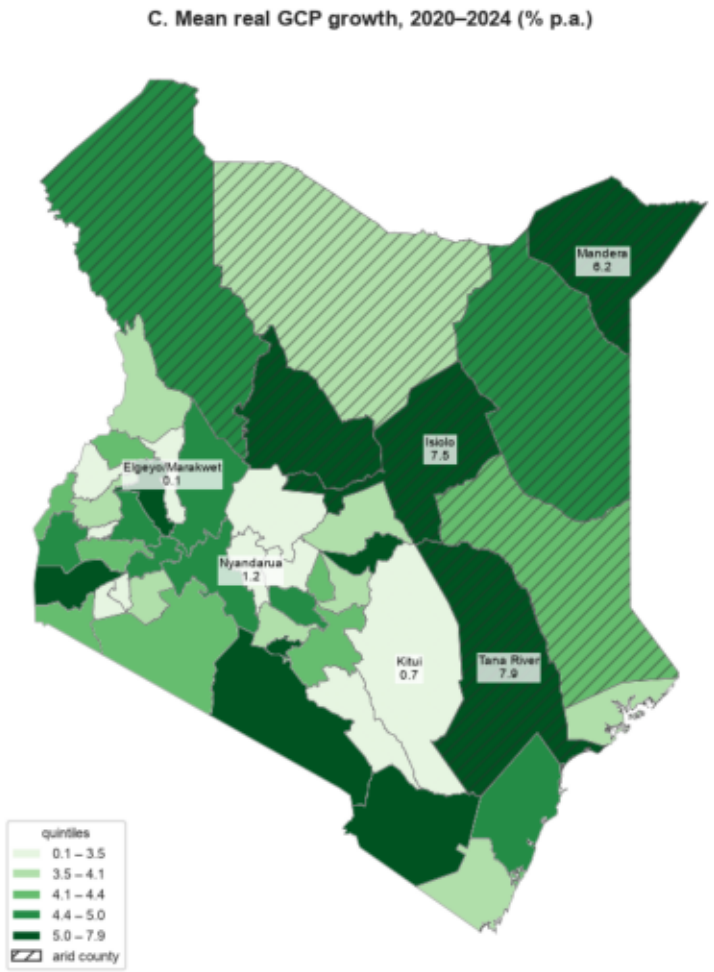
Absorption does not predict growth



- Across 14 specifications — pooled, two-way fixed effects, lagged, extended panel, with/without Nairobi, COVID year or arid counties — the absorption coefficient is small, unstable in sign and never significant.
- Cluster-bootstrap 95 % interval for +10 pp absorption: −0.96 to +0.45 pp of growth.
- A tuned neural network and random forest, evaluated on held-out counties, do not beat a linear model — and assign absorption negligible importance. No hidden threshold.

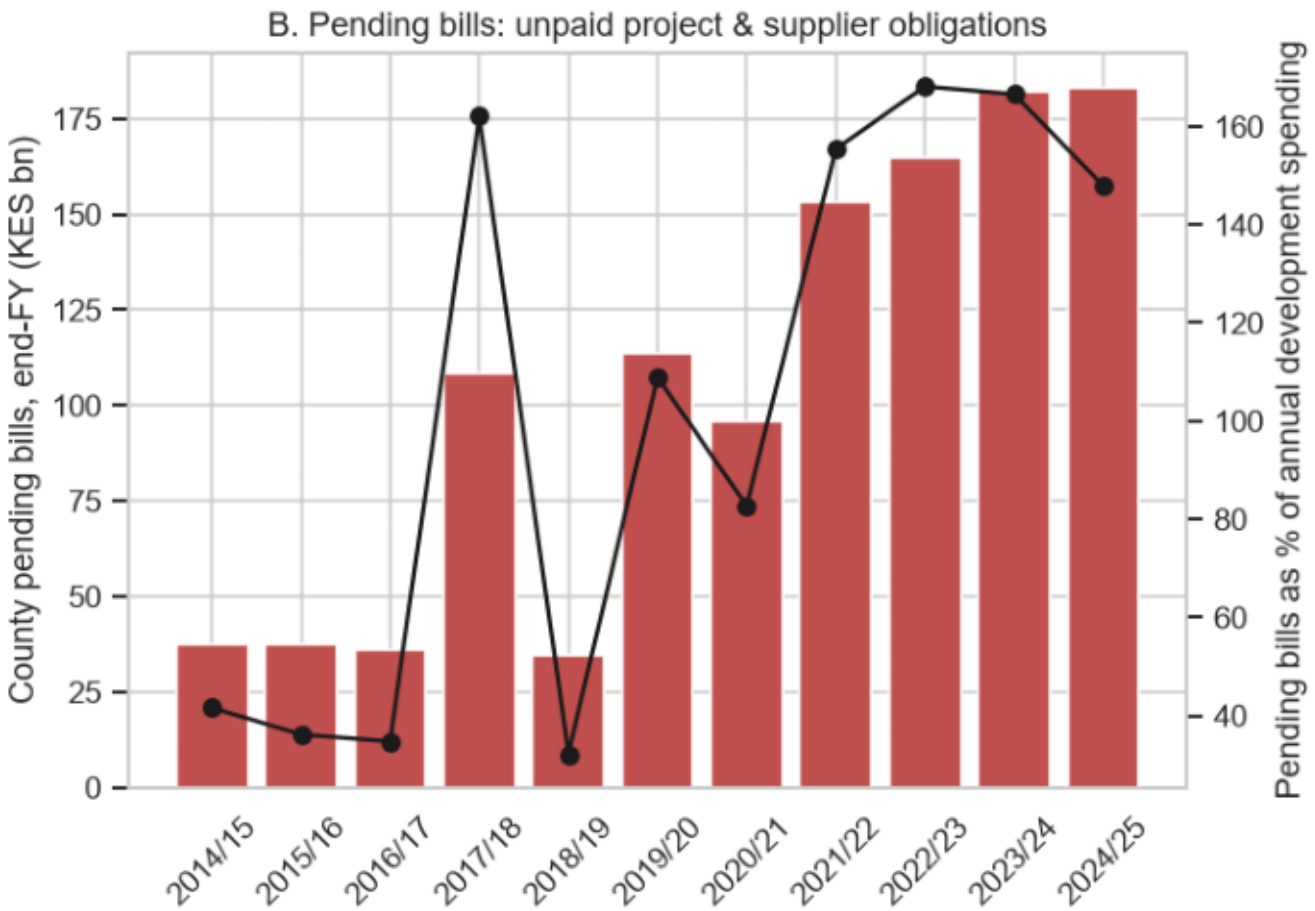
Real GCP: KNBS Gross County Product 2019–2025 editions. SEs clustered by county.

What matters: how much is spent, with a lag



- A 10 % increase in development spending per person is followed by ~0.4 pp faster real GCP growth the next year ($p < 0.01$).
- The eight arid counties — the largest per-capita recipients under the CRA formula — grew 1.9 pp/yr faster than other counties in 2020–24 and saw poverty rise less (+1.2 pp vs +3.7 pp).
- Devolution's redistributive design is visible in the outcomes. The level of spending carries information; the share of budget executed does not.

Poverty and projects



- Poverty rose in 33 of 47 counties between 2015/16 and 2022. Decade-average absorption has a modest association with smaller increases (−1.8 pp per 10 pp, $p < 0.05$) — but only when averaged over ~9 years, not 3.
- Only 30 % of 762 flagship county projects (FY2022/23) were reported complete. Completion is unrelated to the absorption rate.
- Pending bills: KES 35 bn (2015) → 108 bn (2018 election year) → 183 bn (2025) ≈ 1.5 years of total county development spending. Paying old bills counts as absorption without finishing anything new.

Poverty: KNBS KIHBS 2015/16, Kenya Poverty Report 2022. Projects: COB FY2022/23 county chapters.

SO WHAT?

Three takeaways

- 1 · The absorption rate is a weak proxy for development. It measures cash leaving the treasury relative to a budget whose size is itself a choice — not assets delivered or welfare gained. Judging counties by it alone is weakly founded, in either direction.

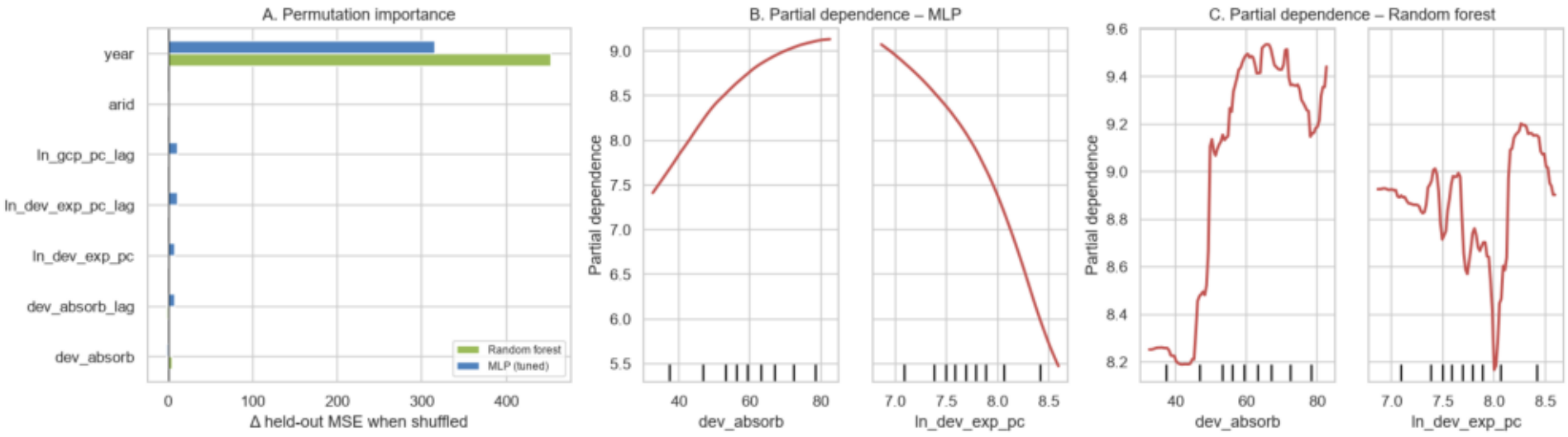
- 2 · Devolution has redistributed. The historically marginalised counties received the most per person, grew fastest and resisted the poverty shock best. Spending levels, with a one-year lag, are associated with growth.

- 3 · Execution matters cumulatively. A decade of consistent budget execution shows up modestly in poverty; a single good year does not. Multi-year completion records would be a more informative yardstick than annual absorption.

- Implications worth testing: report completed projects and unit costs alongside absorption; track the realised (not budgeted) development share; address the KES 183 bn pending-bills stock; smooth exchequer release timing.

UNDER THE HOOD

Methods, data & full analysis



- Data: 12 COB annual reports (6 recovered via the Wayback Machine), CRA CARA schedules, Treasury exchequer releases, KNBS GCP 2013–2024, KIHBS 2015/16 & KCHS 2022, census 2009/2019, geoBoundaries.
- Methods: two-way fixed-effects panel (linearmodels), long-difference cross-sections, 400-draw county cluster bootstrap, county-grouped cross-validated MLP / random forest / ridge benchmark, choropleths (geopandas).
- Every number traces to a cited official table; parse-inconsistent rows were dropped, not imputed. Full notebook, HTML report, data and provenance logs: github.com/elmanpeter/econometrics-in-action
- Feedback, corrections and collaboration welcome — especially from anyone with county-level project completion data.